Schur-functor representations of $U(n)$

Exact trace identities, the holomorphic obstruction, and reproducible matrix tests

Proof and verification note

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Outcome. The finite-dimensional plethystic trace formula is correct. For five pairs (n, λ), including $\lambda = (2, 1)$, 40 direct Schur-matrix comparisons pass with maximum error 2.98×10^{-15}. The important qualification is exact: for nonempty λ, the purely holomorphic $U(n)$ sigma-MGF is 1. The nontrivial statement is the joint law with the conjugate. Three seeded Haar diagnostics also pass.

1 Statement with the unitary correction

Let $W_n = \mathbb{C}^n$, let $U \in U(n)$, fix $\lambda \vdash d$, and put $V_{\lambda,n} = S_\lambda(W_n)$. If $X(U)$ is the eigenvalue alphabet of U, then the eigenvalue alphabet of $S_\lambda(U)$ is the plethysm $s_\lambda[X(U)]$. Hence, for every $r \geq 1$,

$$\boxed{\operatorname{Tr}(S_\lambda(U)^r) = p_r[s_\lambda][X(U)] = s_\lambda[X(U^r)].} \tag{1}$$

For a partition τ, the coefficient integrand of the sigma-MGF is

$$\prod_i h_{\tau_i}[s_\lambda][X(U)]. \tag{2}$$

Associativity of plethysm gives $h_\tau[s_\lambda[X]] = (h_\tau[s_\lambda])[X]$, so Haar integration yields

$$[m_\tau]\mathcal{S}^U_{\lambda,n} = \dim \left(\bigotimes_i \operatorname{Sym}^{\tau_i} S_\lambda(W_n) \right)^{U(n)}. \tag{3}$$

Proposition (central-character obstruction). If $d > 0$, then

$$\boxed{\mathcal{S}^U_{\lambda,n} = 1.} \tag{4}$$

Indeed, zI acts on $S_\lambda(W_n)$ by z^d and on the tensor product above by $z^{d|\tau|}$. For $|\tau| > 0$ this character is not identically one on $U(1)$, so its fixed space is zero. This is an exact finite-n proof, not an asymptotic statement.

The nontrivial object uses a conjugate alphabet. In the stable range its coefficient is

$$[m_\tau \bar{m}_\nu]\mathcal{J}^U_{\lambda,\infty} = \left\langle \operatorname{Exp}_\sigma(h_1\bar{h}_1),\, h_\tau[s_\lambda]\, \overline{h_\nu[s_\lambda]} \right\rangle. \tag{5}$$

This is the plethystic pushforward of the defining-representation joint law.

Stable-range qualification. Let $|\lambda| = d$ and consider a balanced coefficient indexed by outer partitions α, β, with $d|\alpha| = d|\beta| =: m$. Embed its coefficient space into $W_n^{\otimes m} \otimes (W_n^*)^{\otimes m}$. The first fundamental theorem for GL_n says that contraction tensors indexed by S_m span the invariants, and the second fundamental theorem says that they are linearly independent when $n \geq m$. Therefore the stable plethystic formula is valid whenever

$$n \geq m = d|\alpha| = d|\beta|.$$

If the two degrees are unequal, the coefficient is exactly zero for every n by the scalar-center argument.

2 Explicit polynomial-Gaussian law

The Frobenius formula gives

$$s_\lambda = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} p_\alpha, \qquad z_\alpha = \prod_j j^{m_j(\alpha)} m_j(\alpha)!. \tag{6}$$

Substituting $p_k[X(U^r)] = \mathrm{Tr}(U^{rk})$ in (**??**) gives the exact identity

$$\mathrm{Tr}(S_\lambda(U)^r) = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} \mathrm{Tr}(U^{ra}). \tag{7}$$

For fixed powers, $\mathrm{Tr}(U^k)$ converges jointly in moments to independent circular complex Gaussians Z_k satisfying $\mathbb{E}Z_k = \mathbb{E}Z_k^2 = 0$ and $\mathbb{E}|Z_k|^2 = k$. Polynomial substitution in (**??**) therefore gives

$$\mathrm{Tr}(S_\lambda(U)^r) \longrightarrow Q^U_{\lambda,r} = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} Z_{ra}. \tag{8}$$

For example, $Q^U_{(2),1} = \frac{1}{2}(Z_1^2 + Z_2)$ and $Q^U_{(1,1),1} = \frac{1}{2}(Z_1^2 - Z_2)$. Pure holomorphic moments vanish, while mixed moments generally do not.

3 Independent finite-dimensional verification

For each test the code draws a seeded Haar unitary by complex Gaussian QR factorization. It constructs a standard Young symmetrizer c_λ on $W_n^{\otimes d}$, obtains an orthonormal basis B of its image by singular-value decomposition, and forms the actual matrix

$$D_\lambda(U) = B^* U^{\otimes d} B. \tag{9}$$

This route does not use symmetric-function trace identities. The comparison route computes $p_k = \mathrm{Tr}(U^{rk})$, obtains h_j from Newton's recurrence

$$j h_j = \sum_{k=1}^{j} p_k h_{j-k}, \qquad h_0 = 1, \tag{10}$$

and evaluates $s_\lambda = \det(h_{\lambda_i - i + j})$ by Jacobi–Trudi. It compares both (**??**) for $r = 1, 2, 3$ and (**??**) for $\tau = (), (1), (2), (1,1), (3)$.

n	tested λ	direct checks	largest purpose	status
2	$(1), (2)$	trace and sigma integrands	low-rank boundary	PASS
3	$(1, 1), (3), (2, 1)$	trace and sigma integrands	mixed Young symmetry	PASS
			40 comparisons; maximum absolute error	2.98×10^{-15}

The seeded $U(8)$ Haar run uses 1,200 matrices. It estimates $\mathbb{E}\chi_{(2)} = 0.0222 + 0.0343i$, $\mathbb{E}\chi_{(2)}^2 = 0.0322 - 0.0404i$, and $\mathbb{E}\chi_{(1,1)} = -0.0059 + 0.0148i$, all within 4.5 estimated standard errors of the exact target zero. These statistical rows check the implementation but do not replace the central-character proof.

4 Scope and reproducibility

The exact trace formula holds whenever $S_\lambda(\mathbb{C}^n)$ is defined ($\ell(\lambda) \leq n$). Coefficient stabilization is a separate assertion: the balanced unitary threshold is $n \geq d|\alpha| = d|\beta|$, while the orthogonal/symplectic Brauer bound is stated in their respective notes. Unitary holomorphic vanishing is exact for every allowed n. Run the verification module below from the repository root, or open `marimo_notebooks/schur_unitary.py` with marimo.

```
python -m lambda_distributions.proofs.schur_functor_classical_groups.unitary.verification
```

References

S. Howe, *Random matrix statistics and zeroes of L-functions via probability in λ-rings*.
G. Lehrer and R. Zhang, *The Brauer Category and Invariant Theory*.